Role of environmental estrogens in the deterioration of male factor fertility.

OBJECTIVE: To evaluate the role of the environmental estrogens polychlorinated biphenyls (PCBs) and phthalate esters (PEs) as potential environmental hazards in the deterioration of semen parameters in infertile men without an obvious etiology. DESIGN: Randomized controlled study. SETTING: Tertiary care referral infertility clinic and academic research center. PATIENT(S): Twenty-one infertile men with sperm counts <20 million/mL and/or rapid progressive motility <25% and/or <30% normal forms without evidence of an obvious etiology and 32 control men with normal semen analyses and evidence of conception. Semen and blood samples were obtained as part of the treatment protocol. MAIN OUTCOME MEASURE(S): Evaluation of semen parameters such as ejaculate volume, sperm count, motility, morphology, vitality, osmoregulatory capacity, sperm chromatin stability, and sperm nuclear DNA integrity. RESULT(S): PCBs were detected in the seminal plasma of infertile men but not in controls, and the concentration of PEs was significantly higher in infertile men compared with controls. Ejaculate volume, sperm count, progressive motility, normal morphology, and fertilizing capacity were significantly lower in infertile men compared with controls. The highest average PCB and PE concentrations were found in urban fish eaters, followed by rural fish eaters, urban vegetarians, and rural vegetarians. The total motile sperm counts in infertile men were inversely proportional to their xenoestrogen concentrations and were significantly lower than those in the respective controls. CONCLUSION(S): PCBs and PEs may be instrumental in the deterioration of semen quality in infertile men without an obvious etiology.